

Arnav Sagar

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Education

- **Delhi Technological University (DTU), Delhi** New Delhi, India
B.Tech. – Software Engineering, CGPA: 8.76 Jul 2024 – Jun 2028
- **Kendriya Vidyalaya Sector-47, Chandigarh** Chandigarh, India
12th: 93% 10th: 94.4% 2020 – 2023

Work Experience

- **Regavis Labs Pvt Ltd** New Delhi, India
Machine Learning Intern [Certificate] Jun 2026 – Present
 - Architected a scalable two-stage cascade architecture for real-time audio deepfake detection using SSL-based verification, LFCC-LCNN screening and prompt tuned **XLSR + AASIST** and to reduce compute cost by **85%** for live calling systems.
- **5G Lab(Delhi Technological University), Department of Telecommunications, GoI** New Delhi, India
Research Intern (5G, MEC, URLLC) [GitHub] [Certificate] Jun 2025
 - Engineered a **sub-25ms P95** edge inference pipeline via per-channel entropy calibration, achieving **4x model compression** and **1.8× throughput** at <1.5% mAP loss, deployed across **CUDA, Metal MPS, and CPU** backends.
 - Built a real-time multi-person tracker using Lucas-Kanade optical flow with IoU-based keypoint association, feeding a **self-supervised trajectory autoencoder** (~1,600 params, <0.1ms/track) over a simulated **5G URLLC transport layer** (2ms latency, 0.1% packet loss); validated by a **29**-test suite.
- **AIMS-DTU** New Delhi, India
Research Intern (LLM Safety) [GitHub] [API] [Certificate] Jun 2025 – Jul 2025
 - Deployed a *session-aware* jailbreak & toxicity classifier with MiniLM cosine similarity, **SHAP** token attribution, adversarial normalization (homoglyph, zero-width, base64), **Redis** caching and an **ONNX**-optimized inference.

Relevant Projects

- **CLASP — Claude API Switching Proxy** [GitHub]
 - Architected a rate-limit-aware reverse proxy routing LLM inference traffic across 18 providers using pre-emptive token-bucket rate limiting, multi-key pool rotation, and circuit breakers, with a fast-path layer resolving trivial probe requests locally and an **SSRF-safe** web-fetch tool using DNS-pinned IP validation.
 - Built an async priority queue with SSE keep-alive absorption to buffer requests during quota exhaustion, backed by a three-tier cache (LRU→SQLite→FAISS) and a cross-provider translation layer; validated with **934** automated tests.
- **CAF-OTSRNet — Thermal Super-Resolution Model** [GitHub] [Demo]
 - Developed a **triple encoder cross-attention** fusion architecture for thermal super-resolution with two-stage optical-thermal alignment module and a **progressive Laplacian decoder** for multi scale reconstruction.
 - Integrated texture guided safety constraints and **physics aware losses** to suppress hallucinations, improving PSNR by **16%** and SSIM by **8.22%** compared to *SOTA model* as evaluated on the dataset provided by **ISRO**.
- **Retrieval Augmentation System** [GitHub]
 - Built a multi-stage RAG system processing 15k+ PDF pages with semantic chunking, hierarchical indexing, **HyDE-based query expansion**, and query decomposition, improving answer relevance by **38%** (measured via GPT-4 judge).
 - Combined hybrid dense-sparse retrieval (**FAISS + BM25**) with RRF fusion, cross-encoder reranking, **MMR** diversity selection, contextual compression, and **CRAG**-based hallucination suppression, reducing hallucination rate by **47%**.
- **Text Splitting and Embedding Visualizer** [GitHub] [Demo]
 - Created a plugin-based visualization tool in vanilla JS for analyzing RAG chunking strategies, using an extensible splitter registry (character, token, sentence, recursive, language-aware) to compute chunk metrics and a cosine similarity matrix.
- **Hybrid CTC-Attention Scene Text Recognition** [GitHub]
 - Designed a **CRNN** architecture that features a VGG-style CNN backbone, combining a 2-layer **BiLSTM** encoder with *Bahdanau attention* decoder for sequence modeling.
 - Implemented a joint CTC and Cross-Entropy loss with **beam search** to stabilize early training and refine character alignment, achieving **89%** word accuracy on the **IIIT5K** benchmark after training on **9M** images.

Technical Skills

- **Languages:** Python, C/C++, JavaScript, HTML/CSS, SQL
- **Developer Tools & Backend:** REST APIs, AsyncIO, Redis, WebSockets, Docker, FastAPI, Weights & Biases, Git, Streamlit
- **ML/AI Frameworks:** PyTorch, TensorFlow, ONNX, HuggingFace, Transformers, OpenCV, Pandas, NumPy, LlamaIndex

Achievements

- National Finalist, Smart India Hackathon 2025 Ahmedabad, India
- Presented "AI-Based Intrusion Detection Systems in 5G Networks" at PEC Chandigarh, showcasing a completely serverless, on-device inference pipeline that achieved **99.3%** bandwidth reduction via edge-based MEC architecture.

Positions of Responsibility

- Coordinator, Business Bulls, DTU (Finance & Strategy Club)